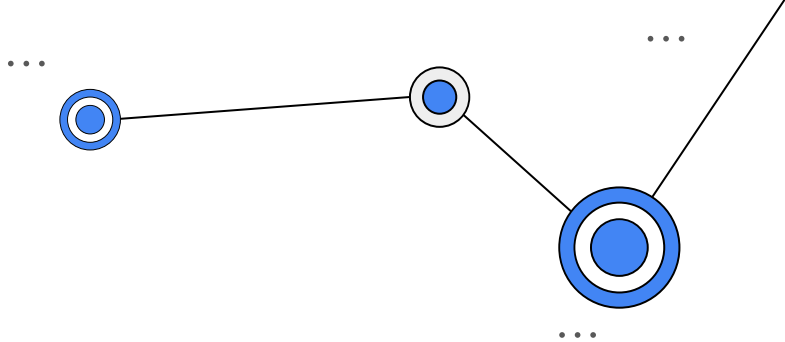
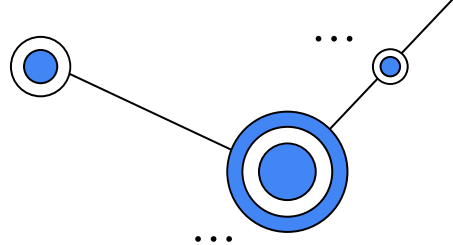




Customer Churn

Classification Using Machine Learning vs.
Deep Learning Neural Network

Random Forest ML Classification

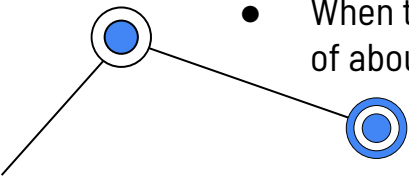


What We Did:

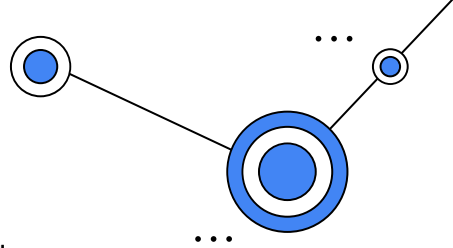
- Separated the independent variables (age, FICO score, balance, etc.) from the dependent variable (whether the customer churned)
- Transformed categorical independent variables into binary variables using one-hot encoding
- Split the dataset into a *training* set (80% of the data) and a *testing* set (20%)
- Trained a random forest classification model on the *training* set, then evaluated its performance on the *testing* set to assess its accuracy
- Trained the model and evaluated how well it generalizes using five-fold cross-validation

Our results:

- The random forest classification model achieved approximately 90.54% accuracy
- When tested on data it hadn't seen before, the model achieved a mean cross-validation score of about 0.9017 (90.17% accuracy)



Deep Learning NN Classification

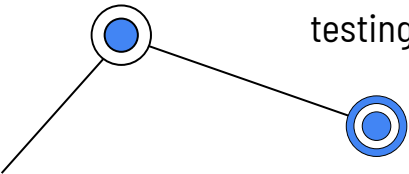


What We Did:

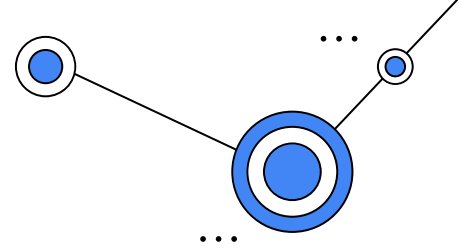
- Prepared the data similar to before (separated independent/dependent variables, one-hot encoded categorical features, scaled numerical features, split data 80/20 into training/testing sets)
- Designed a deep learning model consisting of a 7-layer neural network:
 - 3 trainable linear layers reducing neuron count from 64 at input layer to 2 at output
 - ReLU activation layers after each linear layer to capture non-linear patterns
 - Dropout layers after each activation layer to mitigate overfitting
- Optimized and trained the model using Adam optimization and binary cross-entropy loss

Our Results:

- After 50 epochs of training, the model achieved approximately 91.14% accuracy on the testing set



Comparison, Applications, and Future Plans



- Compare and contrast machine learning and neural networks:
 - Both achieved 90-93% accuracy – they were able to find the correct weights or fit curve for the data
 - The data is more structured with little dimensionality, which explains this
- With data like raw, unstructured text or audio about customer information, it is better to use deep learning because NNs can automatically learn complex unstructured patterns that traditional ML cannot without feature engineering
- To better achieve our goal in predicting customer churn, we plan on collecting more data sources and addressing algorithmic bias
 - If we had data like the conversations between the clients and our company representatives, a deep learning model would likely be a better choice to recognize churn probability based on tone

